(i.e. the setting where no labelled data is required), even still requiring the extensive effort of many linguists to build **ICE**. We point to a more general trend of how to improve not just the coverage of languages, especially across typologically diverse languages, but to improve the cultural sensitivity of language model and language technology evaluation (Hershcovich et al., 2022).

10.2 Missing metrics

As we enumerate in Table 3, there are numerous desiderata that we could have for AI systems and language technologies that we do not currently evaluate. In large part, what we currently measure in HELM reflects what is feasible given the the information and access we have to language models. Further, given language models may be integrated in broader systems, we do not currently taxonomize nor evaluate desiderata (e.g. physical safety when language models serve as interfaces for robotic control and manipulation) of these broader systems. For the desiderata we do taxonomize, beyond having to improve model access to adequately measure them, we specifically highlight (i) user experience, given models increasingly function as user interfaces (see Lee et al., Forthcoming), (ii) linguistic plausibility, given models show language capabilities that invite comparison with humans (see Linzen, 2020), and (iii) provenance/credibility, given current modeling approaches often forsake provenance in favor of other desiderata (see Metzler et al., 2021; Khattab et al., 2021).

For the metrics we do measure, we highlight the following specific axes for improvement. For both robustness and fairness when using perturbations, appropriate estimation of when perturbations should be introduced (see Dhole et al., 2021; Ziems et al., 2022) remains a challenge for constructing synthetic data that is realistic. Further, when using contrast sets to measure robustness equivariance and demographic metadata to measure performance disparities, availability of these resources is the key challenge, especially for more generative scenarios. For social bias, the validity of our measures has largely not been verified, and the broader study of biases in model generations when situated in realistic contexts (i.e. without assuming access to specific metadata) remains quite open. Similarly for toxicity, we resort to using the PerspectiveAPI in spite of its established flaws, and broader protocols that would enable toxicity measurement that reflects the perspectives of different social groups and individuals would be desirable (Gordon et al., 2022). Finally, for measurement of training and inference efficiency, improving the reliable disclosure of necessary information would help ensure the accurate measurement of computational costs, environmental emissions, and various runtimes.

10.3 Missing targeted evaluations

In terms of missing targeted evaluations, we first consider targeted evaluations that we did not evaluate. For model capabilities, we note that linguistic understanding, knowledge, and reasoning correspond to core capabilities studied in the vast literature on human cognitive function. Drawing inspiration from that literature, we note that planning is another standard consideration that we do not explicitly study. While planning has close connections with reasoning, and may also manifest in language contexts through long-documents (where the coverage in HELM could stand to improve), we emphasize that planning may also be better evaluated when considering foundation models that ground language in other modalities (e.g. robotic control). For model harms, we identify other forms of harm that we do not currently evaluate. In terms of malicious use cases (akin to disinformation), we note that language models could be used to automate spam generation and for other forms of fraud, which are not currently widely studied in the NLP community. Further, in terms of unintentional harms (akin to model biases), we highlight that there are many other variants of harms, which differ from but are related to bias, such as dehumanization (Mendelsohn et al., 2020), denigration (Caines et al., 2018), and condescension (Wang & Potts, 2019) in the growing canon of work on the harms of language technologies (Bender et al., 2021; Weidinger et al., 2022; Rauh et al., 2022; Kirk et al., 2022)

In terms of improving the existing targeted evaluations, we highlight a specific axis of improvement for each. For linguistic understanding, we evaluate linguistic phenomena across syntax, semantics, and morphology, but do not evaluate other levels of linguistic abstraction, highlighting pragmatic and discourse as a specific area of focus. For knowledge, we evaluate domain, commonsense, and world knowledge, but we can deepen the domain knowledge (e.g. domain-specific knowledge bases beyond Wikidata) as well as expand to so-

cial and cultural knowledge (e.g. SocialIQA; Sap et al., 2019b). For reasoning, we evaluate many forms of reasoning, but we can broaden the domain-specific reasoning beyond law (i.e. **LegalSupport**) as well as consider more language-centric reasoning (e.g. multi-hop reasoning as in HotPotQA; Yang et al., 2018). For copyright, we highlight the potential for broader evaluation of privacy risk (e.g. personally identifying information), evaluation with knowledge of the training (to better understand claims of memorization), and harms of greater societal (e.g. plagiarism of student essays) or legal (e.g. violation of copyright law) consequence. For disinformation, we highlight the use of trained annotators as important for better characterizing true disinformation risk, along with grounded user studies to observe the effect of machine disinformation generation in advancing specific goals of persuasion and deception influencing human behavior. Finally, for bias and toxicity, we reiterate our general position of pushing evaluation towards more contextually-situated measurement (see Rauh et al., 2022), which is why we measured these as generative harms for all generative scenarios beyond isolated evaluations.

10.4 Missing models

For models, we highlight three categories. First, there are models that we can access that we do not evaluate, largely because they were released very close to the release of this work (e.g. Galactica (Taylor et al., 2022) was released on the same day as this work was released). Noteworthy examples, especially given our findings on instruction-tuning, are Flan-T5 (Chung et al., 2022), Tk-Instruct (Wang et al., 2022c), and BLOOMZ (Muennighoff et al., 2022). There are newer versions of commercial APIs from AI21 Labs and Cohere which we have yet to evaluate. We hope the current exclusion of these models will be merely temporary, and that we will be able to reliably evaluate models that are released openly.

Second, there are models that have been disclosed publicly but we do not have access. Notably, at present, prominent models from DeepMind and Google (e.g. Gopher (Rae et al., 2021), Chinchilla (Hoffmann et al., 2022), LaMDA (Thoppilan et al., 2022), PaLM (Chowdhery et al., 2022), and U-PaLM (Tay et al., 2022b)) fall in this category. Consistent with the recommendations of Liang et al. (2022), we believe research access to publicly benchmark and document these models is necessary, even if the broader practices for model release will differ across model providers. To this end, we recommend patterns of developer-mediated access as potential middlegrounds to ensure these models can be benchmarked transparently as a form of structured model access (Shevlane, 2022).

Third, we recognize that there exist many language models, some which potentially play consequential roles in society by underpinning high impact products and services, that are entirely undisclosed. Given the growing importance of language models, and their potential to undergird many different language technologies without this being obvious to the public, we expect new mechanisms for visibility into algorithm pipelines will be necessary (see Bommasani et al., 2022). We raise this as an important open question for the community: how do we ensure foundation models (including language models) are transparently benchmarked when we do not know they exist and no existing mechanism requires they be disclosed?

10.5 Missing adaptation

With the advances in language models and foundation models, we have witnessed the rise of a sprawling class of different adaptation methods (see Bommasani et al., 2021, §4.3). At present, these include a variety of gradient-free prompting strategies (see Liu et al., 2022c) like chain-of-thoughts (Wei et al., 2022c), parameter-efficient gradient-based methods (see He et al., 2022) like adapters (Houlsby et al., 2019), and full gradient-based fine-tuning. Beyond exploring any of these specific methods, we recommend work also explore how interoperable these methods are across different scenarios, metrics, and models so as to understand how best practices for adaptation should emerge. We also emphasize these methods assume different levels of access to models and may use different resources, which we could imagine tracking as means for fairer comparison (Perez et al., 2021; Bommasani et al., 2021, §4.2). Finally, we encourage work to explore model adaptation more expansively beyond specific machine learning methods, such as new modes of interaction that empower humans and broader forms of adaptation such as continual learning (see Bommasani et al., 2021, §2.5, §4.3).

11 Limitations and future work

To understand how our work is limited, and therefore opportunities for future work, we consider three categories: (i) our results, (ii) our benchmark implementation, and (iii) our underlying benchmark design principles.[84]

11.1 Limitations of results

For our results, we identify three key limitations: (i) the relevance for practical use of language models, (i) the generalizability of the findings, and (iii) the dependence on adaptation decisions.

Relevance for practical use. In practice, language models are used not only for diverse scenarios but in diverse contexts. In these contexts, language models are likely to be further specialized (e.g. fine-tuned on much larger data than 5 examples from the domain of interest). Consequently, our results should not be taken as levelling the universal claim that models that perform well are always desirable and models that perform poorly are always undesirable. For example, GPT-J (6B) does not perform particularly well, but may be suitable in many contexts due to its smaller size and the ease of fine-tuning (e.g. with fewer hardware/compute constraints), so that the model can make use of more in-distribution data. More broadly, we expect the totality of the results we provide are not relevant for every practical use case: we anticipate practitioners should first identify scenarios and metrics pertinent to their use conditions, and then prioritize these scenarios/metrics in interpreting the results of this benchmark.

Generalizability. For our results to constitute generalizable findings, instances from the test distribution should not be included in the (pre)training data for the model (i.e. there should be no train-test contamination). However, as several works have discussed (e.g. Brown et al., 2020; Dodge et al., 2021; Sanh et al., 2021), given that the language models we consider are trained on massive, multi-source, and incompletely-documented data (e.g. text scraped from the Internet), it is difficult to directly determine if they are contaminated. We document all known evidence from prior work of contamination in Appendix G, though we acknowledge the extent to which models are contaminated and this compromises the validity of our results largely remains unclear.

Adaptation. We emphasize that our results, and qualitative trends drawn from the results, depend on the choice and implementation of prompting as the adaptation mechanism. In other words, we should not assume we will see the same trends if models are fine-tuned or if prompts are explicitly optimized (Shin et al., 2020; Zhou et al., 2022). This is compounded by the evidence that we show in §8.2: PROMPTING-ANALYSIS, along with several prior works (e.g. Bach et al., 2022), that model behavior is very sensitive to prompt design. Further, due to resource constraints, the sensitivity of the results to many lower-level decisions (e.g. decoding hyperparameters) was not investigated and remains unclear.

11.2 Limitations of HELM implementation

For our implementation, we see the main limitation as the lack of coverage of what is missing. In fact, beyond what we discuss in §11: LIMITATIONS, we emphasize our holistic approach generalizes to the evaluation of any foundation model: in future work, we can imagine specifying both taxonomies and concrete benchmarks for forms of natural language beyond text (e.g. sign, speech), and data modalities even beyond natural language (e.g. images, code, videos, proteins) as in Tamkin et al. (2021; 2022). More generally, we highlight our assumption of/evidence for the validity and reliability of our scenarios and metrics.

Validity and reliability. Many datasets can instantiate a scenario (i.e. agree in task and domain), but differ enormously in the extend to which results reported on those datasets are useful. In the parlance of measurement modeling (Loevinger, 1957; Messick, 1987; 1988; Jackman, 2008; Liao et al., 2021; Jacobs & Wallach, 2021), we would like results we report to be *valid* (i.e. reflect the underlying construct such as summarization of legal documents). While all the datasets we use were introduced in works with some process

[84] We note that there are other important limitations (e.g. what is missing) that we have already discussed.